

Tutorial No : 6

Tutorial Title: Write a program in C++ to perform following operations on complex numbers Add, Subtract, Multiply, Divide. Use operator overloading for these operations. The objective of this assignment is to learn the concepts of Operator overloading.

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Program:

```
#include <iostream>
using namespace std;

class complex {
    float real, img;

public:
    void accept(float r, float i) {
        real = r;
        img = i;
    }

    void display() {
        if (img >= 0) {
            cout << real << " + " << img << "i";
        } else {
            cout << real << " - " << -img << "i";
        }
    }

    complex operator+(complex c2) {
        complex sum;
        sum.real = real + c2.real;
        sum.img = img + c2.img;
        return sum;
    }

    complex operator-(complex c2) {
        complex sub;
```

```

        sub.real = real - c2.real;
        sub.img = img - c2.img;
        return sub;
    }

```

```

    complex operator/(complex c2) {
        complex multiply;
        multiply.real = real * c2.real;
        multiply.img = img * c2.img;
        return multiply;
    }

```

```

    complex operator*(complex c2) {
        complex division;
        division.real = real / c2.real;
        division.img = img / c2.img;
        return division;
    }
};

```

```

int main() {
    complex c1, c2, c3, c4, c5, c6;
    c1.accept(3, -5);
    c2.accept(9, 20);

    cout << "c1 = ";
    c1.display();
    cout << "\nc2 = ";
    c2.display();
    cout << endl;

    c3 = c1 + c2;
    cout << "the addition of two complex: ";
    c3.display();
    cout << endl;

    c4 = c1 - c2;
    cout << "the subtraction of two complex: ";
    c4.display();
    cout << endl;
}

```

```
c5 = c1 / c2;  
cout << "the multiplication of two complex: ";  
c5.display();  
cout << endl;  
  
c6 = c1 * c2;  
cout << "the division of two complex: ";  
c6.display();  
cout << endl;  
  
return 0;  
}
```

Output:

```
c1 = 3 - 5i  
c2 = 9 + 20i  
the addition of two complex: 12 + 15i  
the subtraction of two complex: -6 - 25i  
the multiplication of two complex: 27 - 100i  
the division of two complex: 0.333333 - 0.25i
```

```
=== Code Execution Successful ===
```